We create chemistry

# Safety data sheet



## 1. Substance/preparation and manufacturer/supplier identification

### Basta SL150

Use: crop protection product, herbicide

Manufacturer/supplier:
BASF Vietnam Co. Ltd.
Level 23, Deutsches Haus, 33 Le Duan,
Ben Nghe Ward, Dist. 1, Ho Chi Minh City, Vietnam
Telephone: +84 28 3824 3833
Telefax number: +84 28 3824 3832
E-mail address: minh-triet.thieu@basf.com

Emergency information:
18001703 (Vietnam)
Telefax number: +84 28 3824 3832
International emergency number:
Telephone: +49 180 2273-112

## 2. Hazard identification

Classification of the substance and mixture:
Acute toxicity: Cat. 4 (oral)
Acute toxicity: Cat. 3 (dermal)
Serious eye damage/eye irritation: Cat. 1
Reproductive toxicity: Cat. 1B (fertility)
Reproductive toxicity: Cat. 1B (unborn child)
Specific target organ toxicity — repeated exposure (Nervous system): Cat. 2
Hazardous to the aquatic environment - chronic: Cat. 3

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H318 | Causes serious eye damage. |
| H311 | Toxic in contact with skin. |
| H302 | Harmful if swallowed. |
| H373 | May cause damage to organs (Nervous system) through prolonged or repeated exposure. |
| H360 | May damage fertility. May damage the unborn child. |
| H412 | Harmful to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read label before use. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P260 | Do not breathe dust/gas/mist/vapours. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P270 | Do not eat, drink or smoke when using this product. |
| P264 | Wash contaminated body parts thoroughly after handling. |
| P280 | Wear protective gloves/clothing/eye protection. |

Precautionary Statements (Response):

| | |
|---|---|
| P310 | Immediately call a POISON CENTER or doctor/physician. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P303 + P352 | IF ON SKIN (or hair): Wash with plenty of soap and water. |
| P361 + P364 | Take off immediately all contaminated clothing and wash it before reuse. |
| P330 | Rinse mouth. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents/container to hazardous or special waste collection point. |

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/information on ingredients

**Chemical nature**

crop protection product, herbicide, Soluble concentrate (SL)

Hazardous ingredients

Butanoic acid, 2-amino-4-(hydroxymethylphosphinyl)-, monoammonium salt

| | |
|---|---|
| Content (W/W): 13.51 % | Acute Tox.: Cat. 4 (Inhalation - vapour) |
| CAS Number: 77182-82-2 | Acute Tox.: Cat. 4 (oral) |
| | Acute Tox.: Cat. 4 (dermal) |
| | Repr.: Cat. 1B (fertility) |
| | Repr.: Cat. 1B (unborn child) |
| | STOT RE: Cat. 2 |

1-methoxypropan-2-ol

| | |
|---|---|
| Content (W/W): 7.5 % | Flam. Liq.: Cat. 3 |
| CAS Number: 107-98-2 | Acute Tox.: Cat. 5 (oral) |
| | STOT SE: Cat. 3 (drowsiness and dizziness) |

Polyethyleneglycolmonoalkylethersulphate, salts
Content (W/W): 42.5 %
CAS Number: 68891-38-3

## 4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing. Symptoms of poisoning may occur even after several hours, continue medical observation for at least 48 hours after the accident.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Do not induce vomiting. Call a poison control center or physician for treatment advice. Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: The most important known symptoms and effects are described in the labelling (see section 2) and/or in section 11., vomiting, diarrhea, abdominal cramps, tremors, hypotony, unconsciousness, coma, convulsions, respiratory disorders, nausea, rapid heart rate, Symptoms may be delayed for several hours.
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote. Medical monitoring for at least 24-48 hours.

## 5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen cyanide, nitrogen oxides, sulfur oxides
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

## 7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep only in the original container. Keep container tightly closed in a cool, well-ventilated place. Keep away from heat. Store protected against freezing. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

## 8. Exposure controls and personal protection

Components with occupational exposure limits

1-methoxypropan-2-ol, 107-98-2;
TWA value 50 ppm (ACGIHTLV)
STEL value 100 ppm (ACGIHTLV)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Suitable chemical resistant safety gloves (EN 374) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

| | |
|---|---|
| Form: | liquid |
| Colour: | blue to green |
| Odour: | pungent |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | approx. 6 - 8<br>(23 °C)<br>(undiluted) |
| Melting point: | approx. 0 °C<br>Information applies to the solvent. |
| Boiling point: | approx. 100 °C<br>Information applies to the solvent. |
| Flash point: | No flash point - Measurement made up to pilot light extinguishes. |
| Evaporation rate: | not applicable |
| Flammability (solid/gas): | not applicable |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Ignition temperature: | 405 °C |
| Thermal decomposition: | > 200 °C |
| Explosion hazard: | not explosive (Directive 92/69/EEC, A.14) |
| Fire promoting properties: | not fire-propagating |
| Vapour pressure: | (20 °C) |
| Density: | approx. 1.11 g/cm3<br>(20 °C) |

Relative vapour density (air):
not applicable

Solubility in water: miscible
(20 °C)
Information on: Butanoic acid, 2-amino-4-(hydroxymethylphosphinyl)-, monoammonium salt
Partitioning coefficient n-octanol/water (log Pow): < 0.1
(22 °C; pH value: 7)
Information on: Polyethyleneglycolmonoalkylethersulphate, sodium salt
Partitioning coefficient n-octanol/water (log Pow): 0.3
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Viscosity, dynamic:
not determined

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

## 10. Stability and Reactivity

Conditions to avoid:
See MSDS section 7 - Handling and storage.

Thermal decomposition: > 200 °C

Substances to avoid:
bases

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
ammonia
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Acute toxicity

Assessment of acute toxicity:
Of pronounced toxicity after short-term skin contact. Of moderate toxicity after single ingestion. Inhalation is not likely in the available physical form. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): 1,730 mg/kg

LC50 rat (by inhalation): 2.97 mg/l 4 h
The test result applies only to the substance transferred into respirable aerosol (particles < 20 µm).

LD50 rat (dermal): 593 mg/kg

### Irritation

Assessment of irritating effects:
May cause severe damage to the eyes. Not irritating to the skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit:

### Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:
modified Buehler test guinea pig: (OECD Guideline 406)

### Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

### Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. In long-term studies in mice in which the substance was given by feed, a carcinogenic effect was not observed.

### Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Assessment of reproduction toxicity:
Causes impairment of fertility in laboratory animals.
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### Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Assessment of teratogenicity:

The substance caused malformations/developmental toxicity in laboratory animals. The substance did not cause malformations in animal studies; however, toxicity to development was observed at doses that were toxic to the parental animals.
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### Specific target organ toxicity (single exposure):

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

### Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Assessment of repeated dose toxicity:
Prolonged or repeated exposure may cause neurological disturbances.
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### Aspiration hazard

No aspiration hazard expected.

### Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Ecotoxicity

Assessment of aquatic toxicity:
Harmful to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) 13.4 mg/l, Oncorhynchus mykiss

Aquatic invertebrates:
EC50 (48 h) 17.8 mg/l, Daphnia magna

Aquatic plants:
EC50 (72 h) 8.72 mg/l (growth rate), Pseudokirchneriella subcapitata

(72 h) 1.25 mg/l, Pseudokirchneriella subcapitata

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Microorganisms/Effect on activated sludge:
EC50 (3 h) > 1,000 mg/l, activated sludge
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### Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Assessment transport between environmental compartments:
Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.
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### Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).
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### Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: glufosinate ammonium (ISO); ammonium 2-amino-4-(hydroxymethylphosphinyl)butyrate
Bioaccumulation potential:
Bioconcentration factor: 0.05 - 0.3 (42 d), Lepomis macrochirus
Does not accumulate in organisms.
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### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Domestic transport:**

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|---|---|
| Packing group: | III |
| ID number: | UN 2902 |
| Transport hazard class(es): | 6.1 |
| Proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (contains GLUFOSINATE AMMONIUM SOLUTION) |

**Sea transport**
IMDG

| | |
|---|---|
| Packing group: | III |
| ID number: | UN 2902 |
| Transport hazard class(es): | 6.1 |
| Marine pollutant: | NO |
| Proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (contains GLUFOSINATE AMMONIUM SOLUTION) |

**Air transport**
IATA/ICAO

| | |
|---|---|
| Packing group: | III |
| ID number: | UN 2902 |
| Transport hazard class(es): | 6.1 |
| Proper shipping name: | PESTICIDE, LIQUID, TOXIC, N.O.S. (contains GLUFOSINATE AMMONIUM SOLUTION) |

## 15. Regulatory Information

Hazard determining component(s) for labelling: GLUFOSINATE AMMONIUM

**Other regulations**

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Recommended use: crop protection product, herbicide

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.